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Semester I (B.Tech) **Er. No..................**

Academic Year: 2021-22

Jaypee University of Engineering & Technology, Guna

T-1 (Odd Semester 2021)

18B11CI111 – Software Development Fundamentals

Maximum Duration: 1 Hour Maximum Marks: 15

Notes:

1. This question paper has 5 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

Marks

Q1. COVID-19 is the disease caused by a new coronavirus called SARS-CoV-2. The most common symptoms of COVID-19 are **[03]**

- Fever
- Dry cough
- Fatigue

A patient with the presence of one or more of the above symptoms will be diagnosed as a suspect.

Based on the information as detailed above, you are required to write an algorithm (in plain statements) to classify a patient as a suspected COVID-19 positive or COVID-19 Negative.

Q2. A decimal number 56782 has five digits. Similarly, a decimal number 4 is made up of only one digit. Considering the above specifications, you are expected to draw a flow-chart, following every rule taught in your SDF classes, for computing the number of digits in a supplied number. **[03]**

Q3. Write a few arguments in favour and criticism of using pseudocodes in a program design. **[02]**

Q4. When the same object is dropped from rest at different heights, the time it takes to touch the ground is also different. A table showing different heights and corresponding time is presented below. You are required to write a program or a pseudo code that displays such a table on the computer screen. Note: The table shown below is left incomplete intentionally. Your program or pseudo code, however, should display the complete table. **[03]**

Table for Q4: Object dropped from different heights and time it takes to reach the ground.

Height (meters)	Time taken to reach the ground (seconds)
5	1
125	5
405	9
.	.
.	.
.	.
515205	321

Q5. Based on your understanding of the number systems, solve the following questions: **[04]**

(a) Determine the base of the numbers in 24 + 17 =40. Provided that all the numbers in the given operation are in the same base.

(b) A man has $(1111)_2$ bananas. $(10)_8$ bananas are stolen from him. How many bananas he is left with? Your answer should be in radix 7 number system.

(c) Represent $(24.25)_{10}$ into the corresponding binary number

(d) Perform the following subtraction operation using 2s complement method:

$$(1011010)_2 - (1010010)_2$$

H = H   5 = 1
H =   125 = 5
405 = 9

1 ~~xx~~ + 5
n −
515205 321.